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Chapter 1

QUBIC: A Bolometric Interferometer (BI)

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Introduction

Observations of the Cosmic Microwave Background (CMB) have become one of the most powerful probes in modern cosmology, providing stringent constraints on the composition, geometry, and evolution of the Universe. In particular, the polarisation of the CMB carries unique information about the physics of the early Universe. Of special interest is the curl-like component of the polarisation field, known as B -modes, which can be generated by primordial tensor perturbations produced during Inflation. The detection of such primordial B -modes would provide compelling evidence for the quantum origin of cosmological fluctuations and offer a direct probe of the energy scale of Inflation. However, measuring B -mode polarisation remains extremely challenging. The expected signal is several orders of magnitude weaker than the temperature anisotropies and is significantly contaminated by astrophysical foregrounds, such as Galactic dust and synchrotron emission, as well as by instrumental systematics. Achieving the required sensitivity and control over these systematics therefore necessitates innovative experimental strategies.

The Q&U Bolometric Interferometer for Cosmology (QUBIC) has been designed to address these challenges by combining the high sensitivity of bolometric detectors with the intrinsic systematic control of interferometric techniques. Unlike traditional imaging experiments, QUBIC directly measures the interference pattern produced by an array of horns. This approach introduces a natural redundancy in the instrument response, which can be exploited through a self-calibration procedure to mitigate instrumental systematics.

In this chapter, we present the QUBIC experiment in detail. We first describe the instrumental concept and its key components, with particular emphasis on the principles of bolometric interferometry. We then discuss the scientific objectives of QUBIC. Finally, we present the main instrumental subsystems of the telescope, along with their respective roles and limitations.

The QUBIC collaboration published a series of eight papers in 2022 describing the instrument in detail. These articles cover the overall project [1], its unique spectral capabilities [2], and comprehensive descriptions of its key instrumental components [3, 4, 5, 6, 7, 8]. They form the foundation of this chapter.

1.1 Telescopes for CMB physics

1.1.1 Classical approach: Imager

Before introducing QUBIC instrument, it is interesting to first present the design of numerous modern CMB instruments, called *direct imaging*. Imagers are long-exposure telescope, directly forming images of the sky on their focal planes, where detectors are located. This design is widely adopted by CMB experiments, such as Planck [9], Bicep2/Keck Array [10] or Simons Observatory [11]. These experiments use thousands of detectors, each observing a small patch of the sky. These detectors observe photons, measuring their total power. CMB polarisation observations need very high sensitivity detectors. The dominant technology used is bolometer, which are equipping nearly every CMB instrument¹. Bolometers are measuring the modification of conductance, due to a rising of temperature of the thermal mass, after interaction with photons in the absorber (see Figure 1.1).

As detectors are observing different regions of the sky, they are completely independent. Meaning that the raw sensitivity of such instrument is directly related with the number of detectors. Indeed, the noise variance of one detector, assuming white uncorrelated noise, is given by : $\sigma \propto 1/\sqrt{t}$, with t the observing time, or equivalently, the number of time samples. That means that for N independent detectors observing simultaneously, the expected noise variance is : $\sigma \propto 1/\sqrt{N_{det}t_{obs}}$. As a consequence, modern CMB experiment are looking at adding the maximum number of detectors possible in their focal plane, representing a large fraction of their budget, the number of detectors increasing with time².

¹Another type of antenna called Kinetic Inductance Detectors (KIDs) are at the center of many R&D projects, in order to replace bolometers in CMB experiment [12, 13]. KIDs are currently used in NIKA2 camera [14] and tolTEC camera [15]

²Bicep2/Keck Array (2015): ~ 3000 detectors, Simons Observatory (2025): ~ 60000 detectors

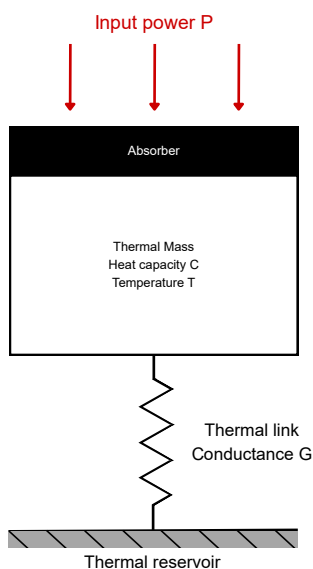


Figure 1.1: Diagram of bolometer. Incoming radiation is interacting in the absorber, in thermal equilibrium with the thermal mass, leading to an increasing of its temperature. This modifies the conductance G of the material, which can be measured, in order to compute the total power of input photons.

1.1.2 Innovative design: Bolometric Interferometry

1.2 QUBIC Experiment

1.2.1 QUBIC Collaboration

1.2.2 Observing Site

1.2.3 Science Objectives

1.3 Instrumental Design

1.3.1 Cryogenics

1.3.2 Filters

1.3.3 Half Wave Plate & Polariser

1.3.4 Horns Array

1.3.5 Optical Combiner

1.3.6 Bolometers

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